

Lesson 3.3. One-Third Fraction of the 3^k Factorial Design

Introduction

The concept of fractional replication can be extended to the 3^k factorial designs. Because a complete replicate of the 3^k design can require a rather large number of runs even for moderate values of k , fractional replication of these designs is of interest. As we shall see, however, some of these designs have complex alias structures.

One-Third Fraction of the 3^k Factorial Design

The largest fraction of the 3^k design is a one-third fraction containing 3^{k-1} runs. Consequently, we refer to this as a 3^{k-1} fractional factorial design. To construct a 3^{k-1} fractional factorial design, select a two-degrees-of-freedom component of interaction (generally, the highest order interaction) and partition the full 3^k design into three blocks. Each of the three resulting blocks is a 3^{k-1} fractional design, and any one of the blocks may be selected for use. If $AB^{\alpha_2}C^{\alpha_3} \dots K^{\alpha_k}$ is the component of interaction used to define the blocks, then $I = AB^{\alpha_2}C^{\alpha_3} \dots K^{\alpha_k}$ is called the **defining relation** of the fractional factorial design. Each main effect or component of interaction estimated from the 3^{k-1} design has two aliases, which maybe found by multiplying the effect by both I and I^2 modulus 3.

As an example, consider a one-third fraction of the 3^3 design. We may select any component of the ABC interaction to construct the design, that is, ABC , AB^2C , ABC^2 , or AB^2C^2 . Thus, there are actually 12 different one-third fractions of the 3^3 design defined by

$$x_1 + \alpha_2 x_2 + \alpha_3 x_3 = u(\text{mod}3)$$

where $\alpha = 1, 2$ and $u = 0, 1, 2$.

Suppose we select the component of AB^2C^2 . Each fraction of the resulting 3^{3-1} design will contain exactly $3^2 = 9$ treatment combinations that must satisfy

$$x_1 + 2x_2 + 2x_3 = u(\text{mod}3)$$

where $u = 0, 1, 2$. The three one-third fractions are as shown below.

Design 1 $u = 0$	Design 1 $u = 1$	Design 1 $u = 2$
000	100	200
012	112	212
101	201	001
202	002	102
021	121	221
110	210	010
122	222	022
211	011	111
220	020	120

If any one of the 3^{3-1} designs above is run, the resulting alias structure is

$$\begin{aligned}
 A &= A(AB^2C^2) = ABC \\
 A &= A(AB^2C^2)^2 = BC \\
 B &= B(AB^2C^2) = AC^2 \\
 B &= B(AB^2C^2)^2 = ABC^2 \\
 C &= C(AB^2C^2) = AB^2 \\
 C &= C(AB^2C^2)^2 = AB^2C \\
 AB &= AB(AB^2C^2) = AC \\
 AB &= AB(AB^2C^2)^2 = BC^2
 \end{aligned}$$

Consequently, the four effects that are actually estimated from the eight degrees of freedom in the design are $A + BC + ABC$, $B + AC^2 + ABC^2$, $C + AB^2 + AB^2C$, and $AB + AC + BC^2$. This design would be of practical value only if all the interactions were small relative to the main effects. Because the main effects are aliased with two-factor interactions, this is a resolution III design. Notice how complex the alias relationships are in this design. Each main effect is aliased with a component of interaction. If, for example, the two-factor interaction BC is large, this will potentially distort the estimate of the main effect of A and make the $AB + AC + BC^2$ effect very difficult to interpret. It is very difficult to see how this design could be useful unless we assume that all interactions are negligible.

The treatment combinations in a 3^{k-1} design with the defining relation $I = AB^{\alpha_2}C^{\alpha_3} \dots K^{\alpha_k}$ can be constructed using a method similar to that employed in the 2^{k-p} series. First, write down the 3^{k-1} runs for a full three-level factorial design in $k - 1$ factors, with the usual 0, 1, 2 notation. Then introduce the k^{th} factor by equating its levels x_k to the appropriate component of the highest order interaction, say $AB^{\alpha_2}C^{\alpha_3} \dots (K - 1)^{\alpha_{k-1}}$ through the relationship

$$x_k = \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_{k-1} x_{k-1}$$

where $\beta_i = (3 - \alpha_k)\alpha_i(\text{mod}3)$ for $1 \leq i \leq k - 1$. This yields a design of the highest possible resolution.

As an illustration, we use this method to generate the design 3_{IV}^{4-1} with the defining relation $I = AB^2CD$. It is easy to verify that the first three digits of each treatment combination in this table are the 27 runs of a full 3^3 design. This is the basic design.

For AB^2CD , we have $\alpha_1 = \alpha_3 = \alpha_4 = 1$ and $\alpha_2 = 2$. This implies that $\beta_1 = (3 - 1)\alpha_1(\text{mod}3) = (3 - 1)(1) = 2$, $\beta_2 = (3 - 1)\alpha_2(\text{mod}3) = (3 - 1)(2) = 4 = 1(\text{mod}3)$, and $\beta_3 = (3 - 1)\alpha_3(\text{mod}3) = (3 - 1)(1) = 2$.

This gives

$$x_4 = 2x_1 + x_2 + 2x_3$$

The levels of the fourth factor satisfy this equation. For example, we have $2(0) + 1(0) + 2(0) = 0$, $2(0) + 1(1) + 2(0) = 1$, $2(1) + 1(1) + 2(0) = 3 = 0(\text{mod}3)$, and so on.

The resulting design has 26 degrees of freedom that may be used to compute the sums of squares for the 13 main effects and components of interactions (and their aliases). The aliases of any effect are found in the usual manner; for example, the aliases of A are $A(AB^2CD) = ABC^2D^2$ and $A(AB^2CD)^2 = BC^2D^2$. One may verify that the four main effects are clear of any two-factor interaction components, but that some two-factor interaction components are aliased with each other. Once again, we notice the complexity of the alias structure. If any two-factor interactions are large, it will likely be very difficult to isolate them with this design.

The statistical analysis of a 3^{k-1} design is accomplished by the usual analysis of variance procedures for factorial experiments. The sums of squares for the components of interaction may be computed as in the previous lessons. Remember when interpreting results that the components of interactions have no practical interpretation.

One-third Fraction of a 3^3 Design: An Example Analysis using R

An experiment was performed to study the effect of three different types of 32-ounce bottles (A) and three different shelf types (B)—smooth permanent shelves, end-aisle displays with gridded shelves, and beverage coolers—on the time it takes to stock ten 12-bottle cases on the shelves. Three workers (factor C) were employed in the experiment, and two replicates of a 3^3 factorial design were run. The observed time data are shown in the following table.

Worker	Bottle Type	Replicate I					Replicate II		
		Per- manent	End Aisle	Cooler			Per- manent	End Aisle	Cooler
1	Plastic	3.45	4.14	5.80	1	Plastic	3.36	4.19	5.23
	28-mm glass	4.07	4.38	5.48		28-mm glass	3.52	4.26	4.85
	38-mm glass	4.20	4.26	5.67		38-mm glass	3.68	4.37	5.58
2	Plastic	4.80	5.22	6.21	2	Plastic	4.40	4.70	5.88
	28-mm glass	4.52	5.15	6.25		28-mm glass	4.44	4.65	6.20
	38-mm glass	4.96	5.17	6.03		38-mm glass	4.39	4.75	6.38
3	Plastic	4.08	3.94	5.14	3	Plastic	3.65	4.08	4.49
	28-mm glass	4.30	4.53	4.99		28-mm glass	4.04	4.08	4.59
	38-mm glass	4.17	4.86	4.85		38-mm glass	3.88	4.48	4.90

Consider the data from replicate I of this experiment. Suppose that only a one-third fraction of this design with $I = ABC$ is run. Construct the design, determine the alias structure, and analyze the data.

```
# Load necessary library
library(DoE.base)

# Create the full design
full_design <- expand.grid(
  Worker = factor(0:2),
  Bottle = factor(0:2),
  Position = factor(0:2)
)

print(full_design)
```

```
##      Worker Bottle Position
## 1         0      0         0
## 2         1      0         0
## 3         2      0         0
## 4         0      1         0
## 5         1      1         0
## 6         2      1         0
## 7         0      2         0
## 8         1      2         0
## 9         2      2         0
## 10        0      0         1
## 11        1      0         1
## 12        2      0         1
## 13        0      1         1
## 14        1      1         1
## 15        2      1         1
## 16        0      2         1
```

```
## 17      1      2      1
## 18      2      2      1
## 19      0      0      2
## 20      1      0      2
## 21      2      0      2
## 22      0      1      2
## 23      1      1      2
## 24      2      1      2
## 25      0      2      2
## 26      1      2      2
## 27      2      2      2
```

```
# Subset to the 1/3 fraction where (W + B + P) mod 3 == 0 (principal fraction)
# We convert factors to numeric for the calculation
fractional_design <- full_design[
  (as.numeric(full_design$Worker)-1 +
   as.numeric(full_design$Bottle)-1 +
   as.numeric(full_design$Position)-1) %% 3 == 0,
]

print(fractional_design)
```

```
##      Worker Bottle Position
## 1         0      0         0
## 6         2      1         0
## 8         1      2         0
## 12        2      0         1
## 14        1      1         1
## 16        0      2         1
## 20        1      0         2
## 22        0      1         2
## 27        2      2         2
```

```
library(tidyverse)
#Y <- c(3.45, 4.30,4.96,3.94,5.15,4.26,6.21,5.48,4.85)
fractional_design <- fractional_design %>%
  mutate(Y = c(3.45, 4.30,4.96,3.94,5.15,4.26,6.21,5.48,4.85))

fractional_design
```

```
##      Worker Bottle Position      Y
## 1         0      0         0 3.45
## 6         2      1         0 4.30
```

```
## 8      1      2      0 4.96
## 12     2      0      1 3.94
## 14     1      1      1 5.15
## 16     0      2      1 4.26
## 20     1      0      2 6.21
## 22     0      1      2 5.48
## 27     2      2      2 4.85
```

```
#Assuming interactions are negligible
model <- lm(Y ~ Worker + Bottle + Position, data = fractional_design)
summary(model)
```

```
##
## Call:
## lm.default(formula = Y ~ Worker + Bottle + Position, data = fractional_design)
##
## Residuals:
##      1      6      8     12     14     16     20     22     27
## -0.25  0.19  0.06  0.06 -0.25  0.19  0.19  0.06 -0.25
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.70000    0.34530  10.715  0.0086 **
## Worker1       1.04333    0.31969   3.264  0.0824 .
## Worker2      -0.03333    0.31969  -0.104  0.9265
## Bottle1       0.44333    0.31969   1.387  0.2999
## Bottle2       0.15667    0.31969   0.490  0.6726
## Position1     0.21333    0.31969   0.667  0.5733
## Position2     1.27667    0.31969   3.993  0.0574 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3915 on 2 degrees of freedom
## Multiple R-squared:  0.9459, Adjusted R-squared:  0.7835
## F-statistic: 5.825 on 6 and 2 DF, p-value: 0.1537
```

```
anova(model)
```

```
## Analysis of Variance Table
##
## Response: Y
##           Df Sum Sq Mean Sq F value Pr(>F)
## Worker     2 2.24887  1.12443   7.3349 0.1200
```

```
## Bottle      2 0.30327 0.15163 0.9891 0.5027
## Position    2 2.80607 1.40303 9.1522 0.0985 .
## Residuals   2 0.30660 0.15330
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The General 3^{k-p} Fractional Factorial Designs

For moderate to large values of k , even further fractionation of the 3^k design is potentially desirable. In general, we may construct a $(\frac{1}{3})^p$ fraction of the 3^k design for $p < k$, where the fraction contains $3k - p$ runs. Such a design is called a $3k - p$ fractional factorial design. Thus, a 3^{k-2} design is a one-ninth fraction, a 3^{k-3} design is a one-twenty-seventh fraction, and so on.

The procedure for constructing a 3^{k-p} fractional factorial design is to select p components of interaction and use these effects to partition the 3^k treatment combinations into 3^p blocks. Each block is then a 3^{k-p} fractional factorial design. The defining relation I of any fraction consists of the p effects initially chosen and their $\frac{3^p-2p-1}{2}$ generalized interactions.

The alias of any main effect or component of interaction is produced by multiplication modulus 3 of the effect by I and I^2 .

We may also generate the runs defining a 3^{k-p} fractional factorial design by first writing down the treatment combinations of a full 3^{k-p} factorial design and then introducing the additional p factors by equating them to components of interaction, as we did before.

We illustrate the procedure by constructing a 3^{4-2} design, that is, a one-ninth fraction of the 3^4 design. Let AB^2C and BCD be the two components of interaction chosen to construct the design. Their generalized interactions are $(AB^2C)(BCD) = AC^2D$ and $(AB^2C)(BCD)^2 = ABD^2$. Thus, the defining relation for this design is $I = AB^2C = BCD = AC^2D = ABD^2$, and the design is of resolution III. The nine treatment combinations in the design are found by writing down a 3^2 design in the factors A and B , and then adding two new factors by setting

$$\begin{aligned}x_3 &= 2x_1 + x_2 \\x_4 &= 2x_2 + 2x_3\end{aligned}$$

This is equivalent to using AB^2C and BCD to partition the full 3^4 design into nine blocks and then selecting one of these blocks as the desired fraction. The principal fraction is given below.

0000	0111	0222
1021	1102	1210
2012	2120	2201

This design has eight degrees of freedom that may be used to estimate four main effects and their aliases. The aliases of any effect may be found by multiplying the effect modulus 3 by AB^2C , BCD , AC^2D , ABD^2 , and their squares. The complete alias structure for the design is given below.

Effect	Aliases							
	I				I^2			
<i>A</i>	ABC^2	$ABCD$	ACD^2	AB^2D	BC^2	$AB^2C^2D^2$	CD^2	BD^2
<i>B</i>	AC	BC^2D^2	ABC^2D	AB^2D^2	ABC	CD	AB^2C^2D	AD^2
<i>C</i>	AB^2C^2	BC^2D	AD	$ABCD^2$	AB^2	BD	ACD	ABC^2D^2
<i>D</i>	AB^2CD	BCD^2	AC^2D^2	AB	AB^2CD^2	BC	AC^2	ABD